6.3 Adding and subtracting: answers

← Unit overview · Practice

For parents: for written sums, look at his working, not only the answer. A wrong answer with a correct method and one slip is a very good sign.

Set 1 · Warm-up

#	Answer	How
1	70	4 tens + 3 tens.
2	700	2 hundreds + 5 hundreds.
3	75	One more ten.
4	50	9 tens − 4 tens.
5	500	8 hundreds − 3 hundreds.
6	236	One more hundred.
7	50	7 + 3 makes a full ten.
8	75	Think of €1 = 100 cents: 100 − 25 = 75.
9	400	50 + 50 = 100.
10	999	One less than 1 000.
11	80	6 tens + 2 tens.

Set 2 · Estimating

Any sensible estimate is fine. These use rounding to the nearest hundred or thousand.

#	Estimate	Exact (for checking)
1	500 + 200 = 700	722
2	300 + 500 = 800	801
3	700 + 200 = 900	903
4	800 − 400 = 400	415
5	3 000 + 2 000 = 5 000	4 930
6	5 000 − 2 000 = 3 000	3 023
7	800	807 (400 + 400 = 800)
8	about €130 (€50 + €80)	€127

Set 3 · Adding column by column

#	Answer	Working (left to right)
1	368	300 + 60 + 8
2	587	500 + 80 + 7
3	722	600 + 110 + 12
4	822	700 + 110 + 12
5	3 621	3 000 + 500 + 110 + 11

#	Answer	Working (left to right)
6	4 202	3 000 + 1 100 + 90 + 12
7	7 175	7 000 + 100 + 70 + 5
8	84 km	46 + 38: 70 + 14 = 84.
9	613 passengers	268 + 345: 500 + 100 + 13 = 613.
10	700	3 hundreds + 4 hundreds.

Set 4 · Subtracting column by column

The working shows the "shortfall" (*tekort*) method. If his school exchanges instead, the answer is the same.

#	Answer	Working (left to right)
1	343	300 + 40 + 3
2	443	400 + 40 + 3
3	169	200 − 30 − 1
4	377	400 − 20 − 3
5	522	1 000 − 500 + 20 + 2
6	3 550	4 000 − 400 − 50
7	€164	489 − 325: 100 + 60 + 4. Or count up from 325 to 489.
8	86 km	214 − 128: 100 − 10 − 4 = 86. Or count up: 128 → 200 is 72, → 214 is 14 more; 72 + 14 = 86.
9	400	9 hundreds − 5 hundreds.

Talk about it: add the answer to the number you took away. You should get the top number back: 169 + 263 = 432.

Set 5 · The written way with exchanging

#	Answer	Key steps (right to left)
1	722	7 + 5 = 12: write 2, carry 1. 8 + 3 + 1 = 12: write 2, carry 1. 4 + 2 + 1 = 7.
2	4 323	6 + 7 = 13 (carry 1); 5 + 6 + 1 = 12 (carry 1); 8 + 4 + 1 = 13 (carry 1); 2 + 1 + 1 = 4.
3	6 027	9 + 8 = 17 (carry 1); 4 + 7 + 1 = 12 (carry 1); 0 + 9 + 1 = 10 (carry 1); 3 + 2 + 1 = 6.
4	169	Exchange a ten: 12 − 3 = 9. Exchange a hundred: 12 − 6 = 6. 3 − 2 = 1.

#	Answer	Key steps (right to left)
5	1 359	5 − 6 can’t be done and there are no tens to exchange, so first exchange a hundred for 10 tens (7 hundreds → 6), then a ten for 10 ones (10 tens → 9). Now 15 − 6 = 9, 9 − 4 = 5, 6 − 3 = 3, and 1 thousand.
6	3 634	No tens or hundreds to exchange, so start at the thousands: 5 thousands → 4 thousands and 10 hundreds → 9 hundreds and 10 tens → 9 tens and 10 ones. Then 12 − 8 = 4, 9 − 6 = 3, 9 − 3 = 6, 4 − 1 = 3.
7	763 bottles	1 250 − 487 = 763. Check: 763 + 487 = 1 250.
8	No: 525	Estimate 800 − 300 = 500, so 675 is much too big. Sam took the smaller digit from the bigger one in the ones (8 − 3) and the tens (7 − 0) instead of exchanging. 803 − 278 = 525.
9	273 seats	1 135 − 862 = 273. Check: 273 + 862 = 1 135.
10	500	Two halves of 500.

Puzzle corner

- **Biggest sum: 1 173.** Put the biggest digits in the hundreds (6 and 5), then 4 and 3 in the tens, then 2 and 1 in the ones: for example 642 + 531 = 1 173. The order inside each column doesn’t matter.
- **Closest to 1 000: 993.** For example 531 + 462 = 993. (Exactly 1 000 isn’t possible with these digits.)
- **Smallest difference: 47.** 412 − 365 = 47. The trick is to make the hundreds digits next to each other (4 and 3), then make the top number’s other digits as small as possible and the bottom number’s as big as possible.

Any sum he makes is a success. The “biggest” question is the most approachable; “closest to 1 000” and “smallest difference” are real challenges.